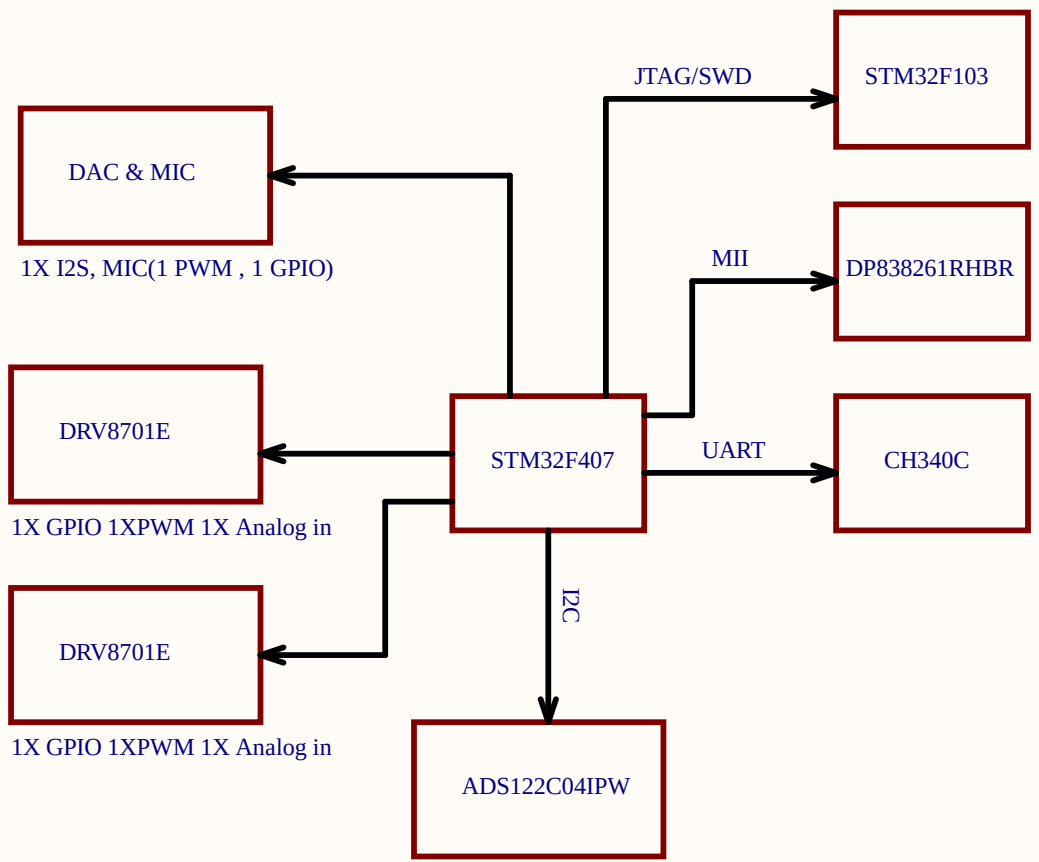
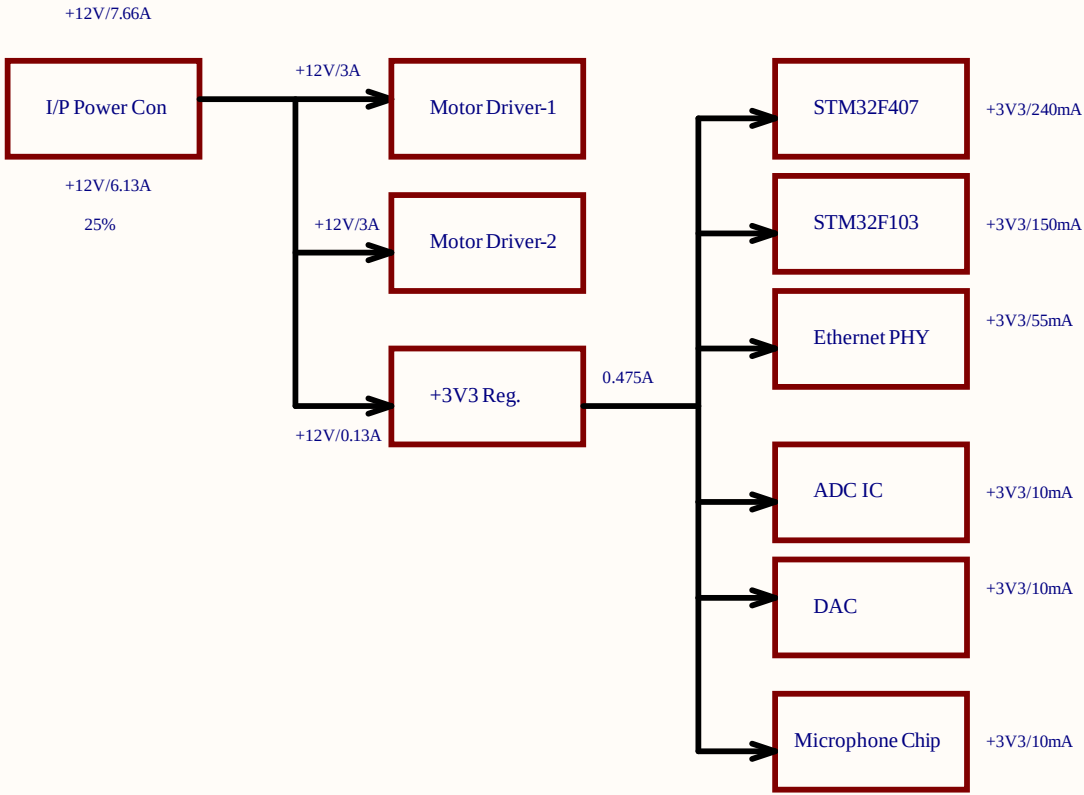


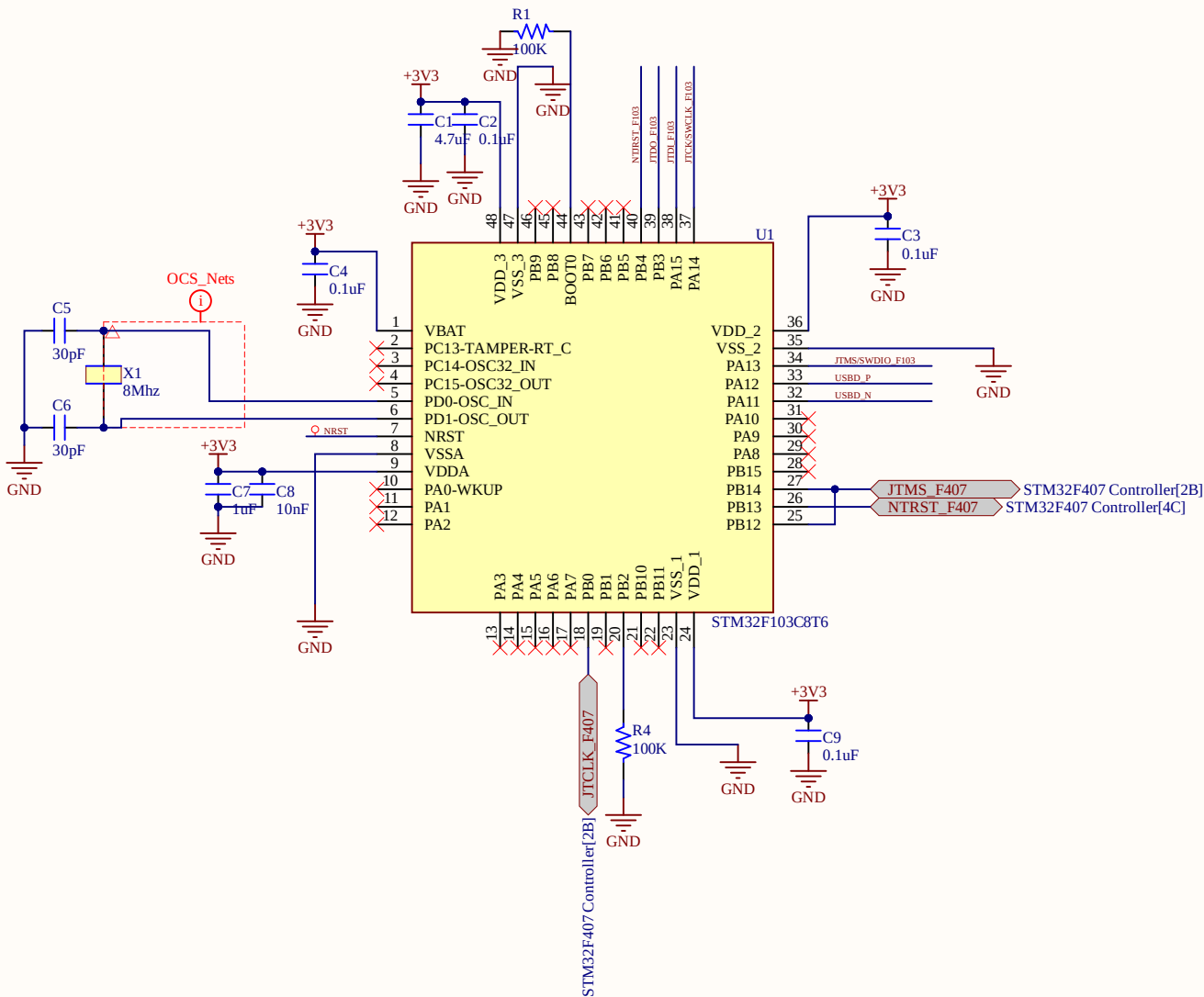




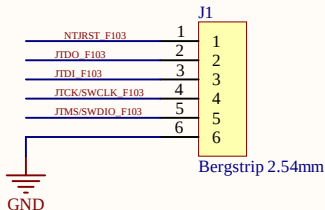
Title STM32F407 Mixed-Signal Hardware Design			
Size: A4	Number:MSD-001	Sheet 2 of 11	Revision: V1.0
Date: 06-10-2026	Time: 21:29:24	Designer Maulik Italiya	
File: D:\Mixed_design_board\My data\Power Budget.SchDoc			

STM32F103 Debugger/Programmer Page

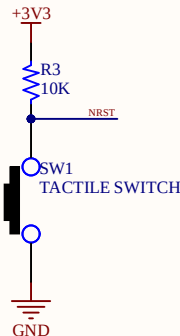
Debugger/Programmer



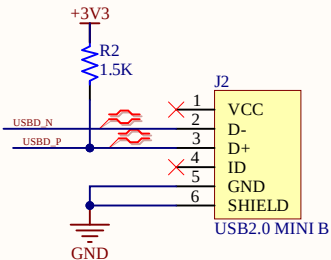
STM32F103 JTAG/SWD



Reset STM32F103



USB2.0 for Program/Debug



Title **STM32F407 Mixed-Signal Hardware Design**

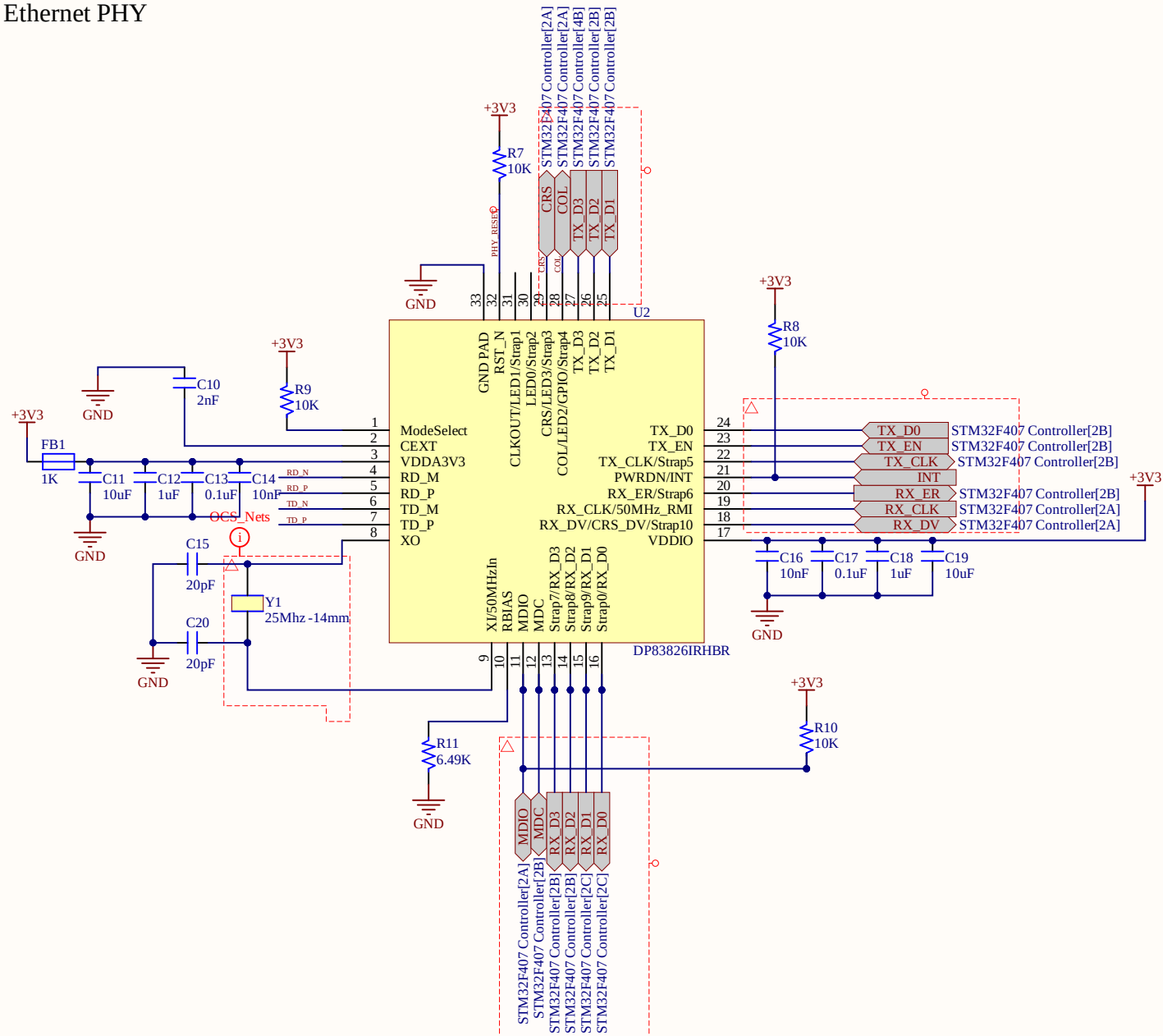
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Date: 06-10-2026 Time: 21:29:24 Designer Maulik Italiya

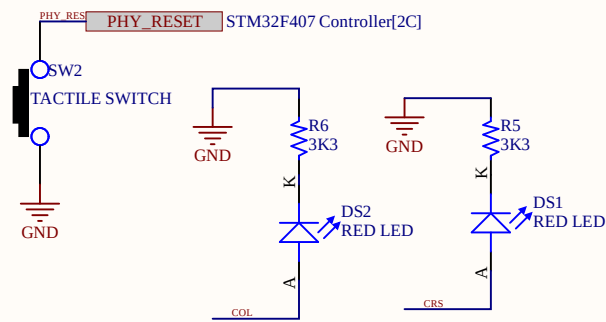
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Ethernet PHY Page

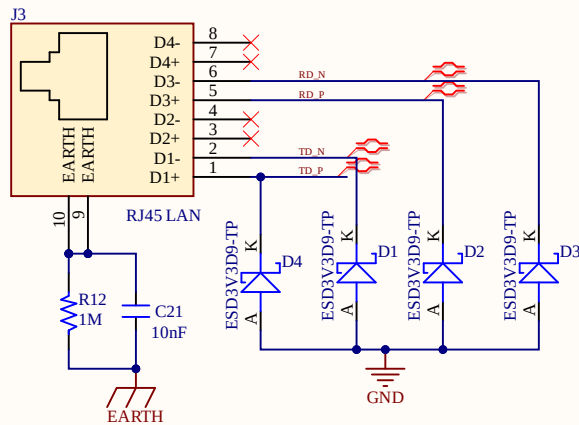
Ethernet PHY



ETH Reset and Fault LEDs



RJ45 Connector



Title **STM32F407 Mixed-Signal Hardware Design**

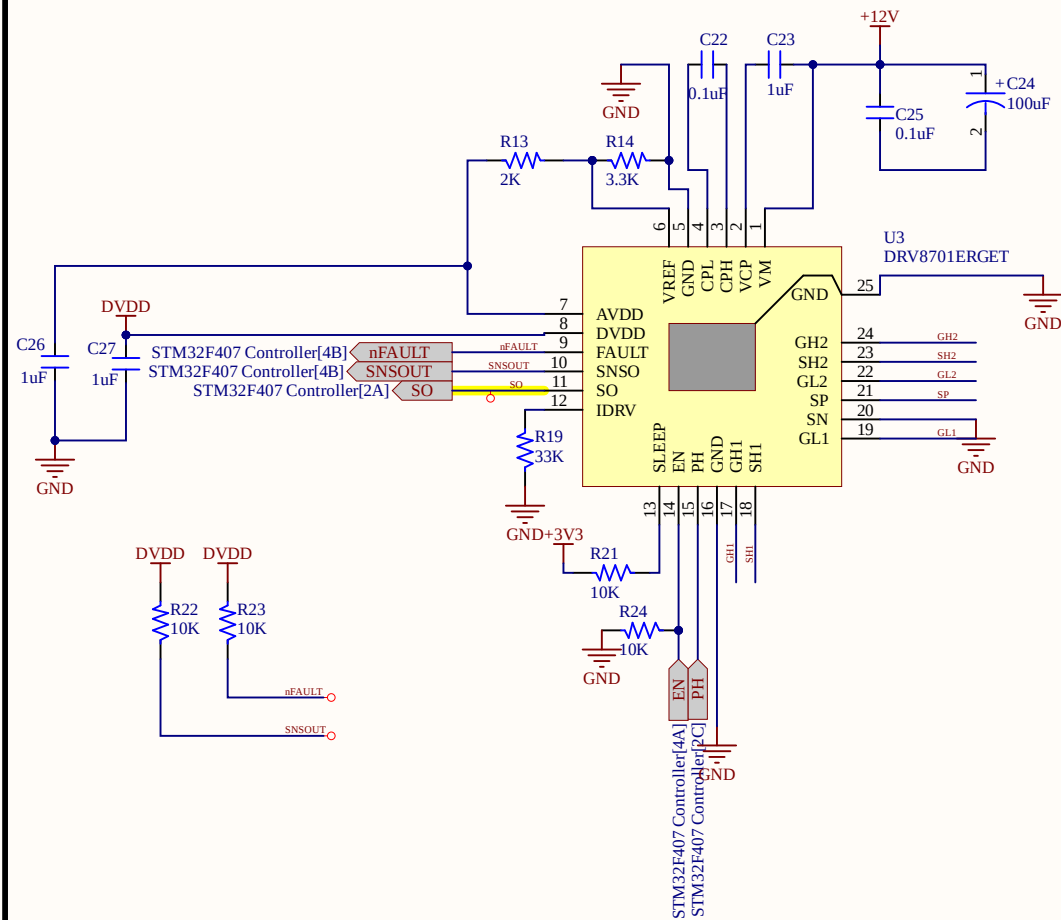
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Date: 06-10-2026 Time: 21:29:24 Designer Maulik Italiya

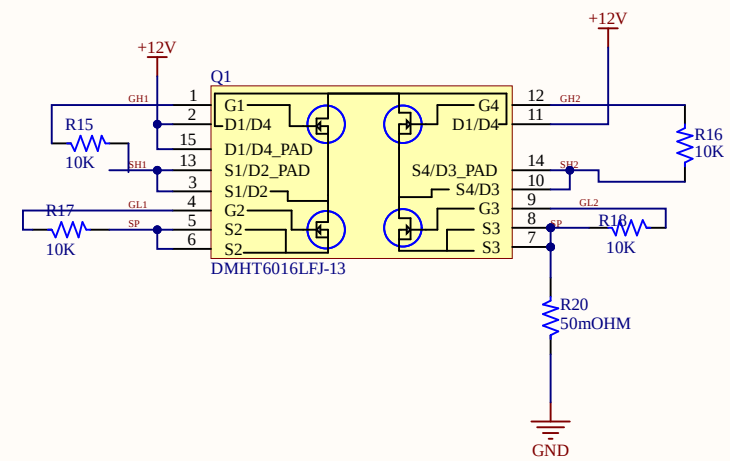
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DRV8701 Motor Driver-1

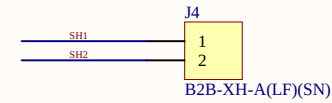
Motor Driver DRV8701



N-MOSFET

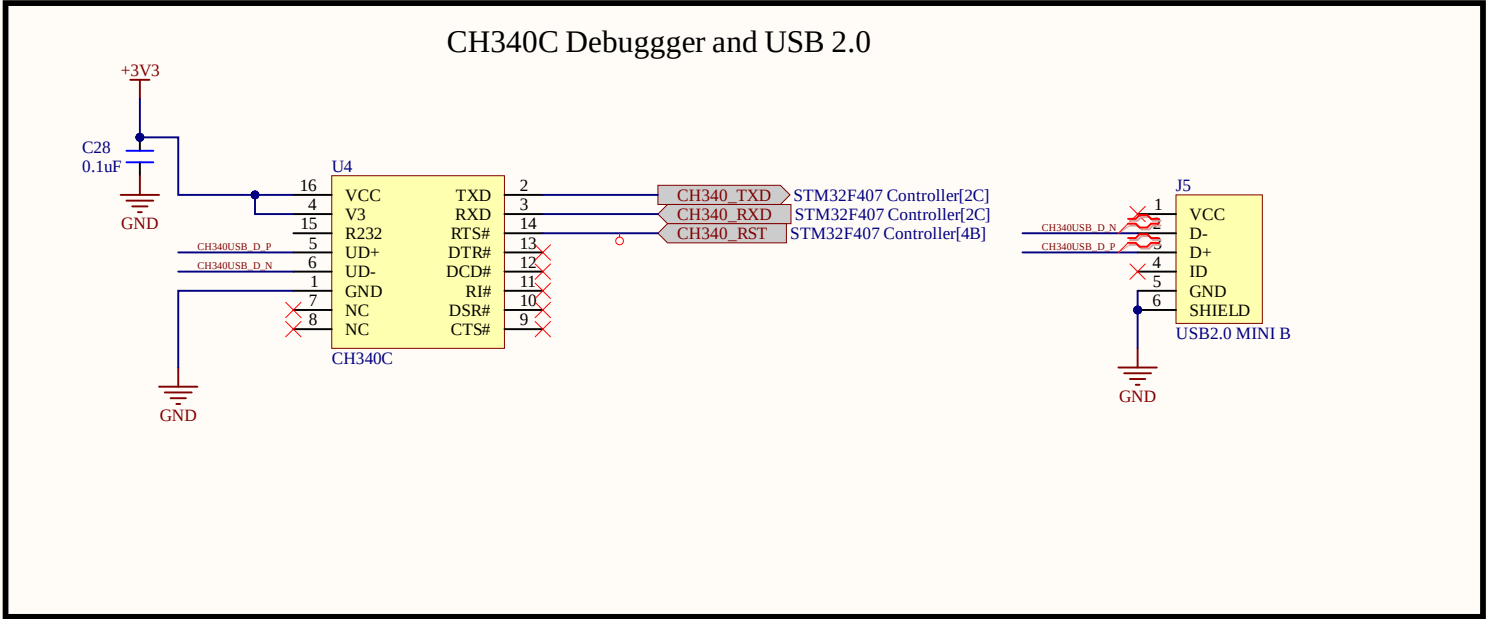


Connector for Motor -1



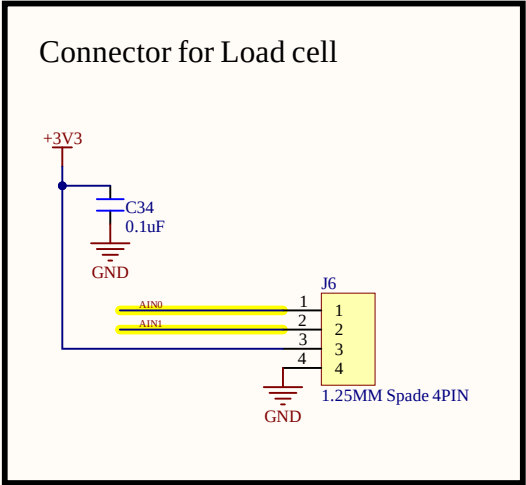
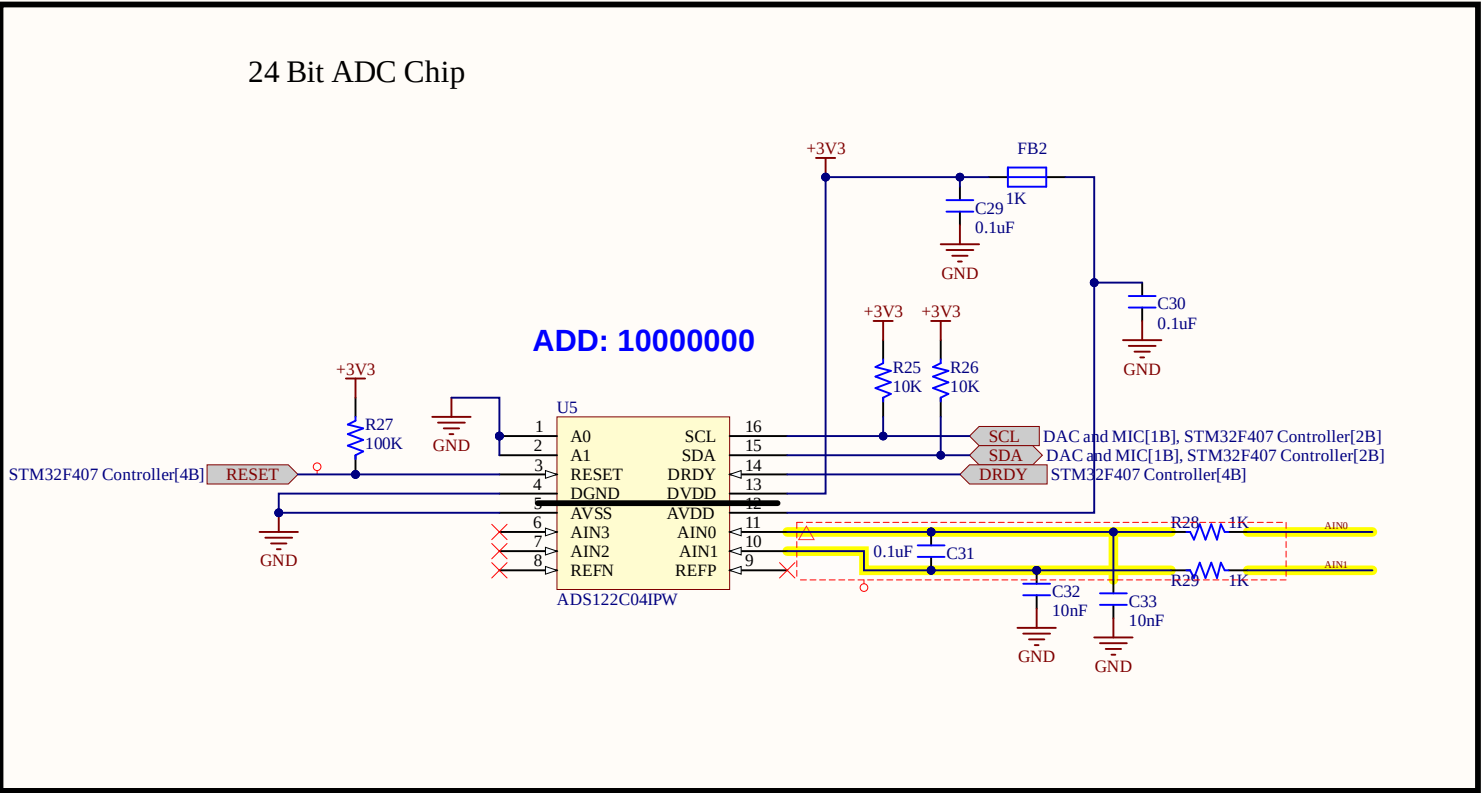
Title <i>STM32F407 Mixed-Signal Hardware Design</i>			
Size: A4	Number: MSD-001	Sheet 5 of 11	Revision: V1.0
Date: 06-10-2026	Time: 21:29:24	Designer Maulik Italiya	
File: D:\Mixed_design_board\My data\Motor Driver Schematic.SchDoc			

CH340C Uart to USB Conv.



Title STM32F407 Mixed-Signal Hardware Design			
Size: A4	Number: MSD-001	Sheet 6 of 11	Revision: V1.0
Date: 06-10-2026	Time: 21:29:24	Designer Maulik Italiya	
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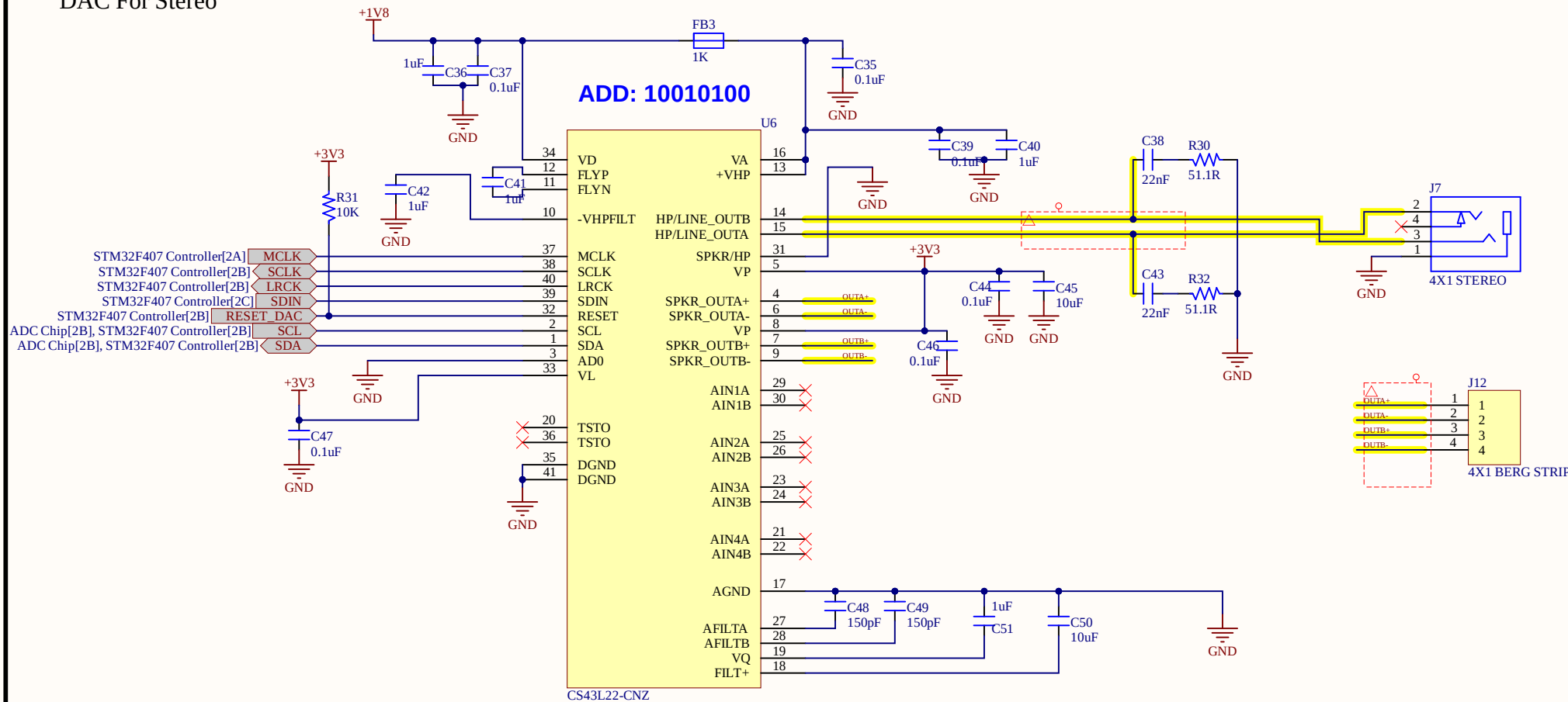
24-Bit ADC for Load cell



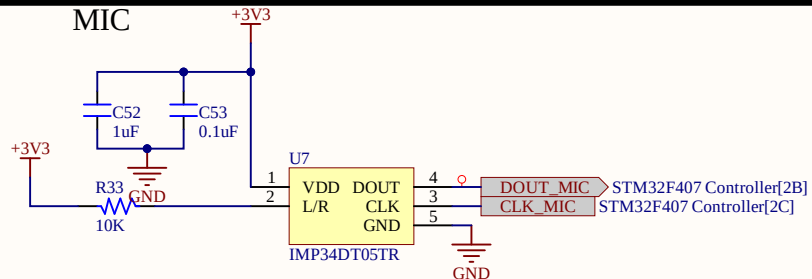
Title STM32F407 Mixed-Signal Hardware Design			
Size: A4	Number: MSD-001	Sheet 7 of 11	Revision: V1.0
Date: 06-10-2026	Time: 21:29:24	Designer Maulik Italiya	
File: D:\Mixed_design_board\My data\ADC Chip.SchDoc			

Stereo DAC and Microphone

DAC For Stereo



MIC



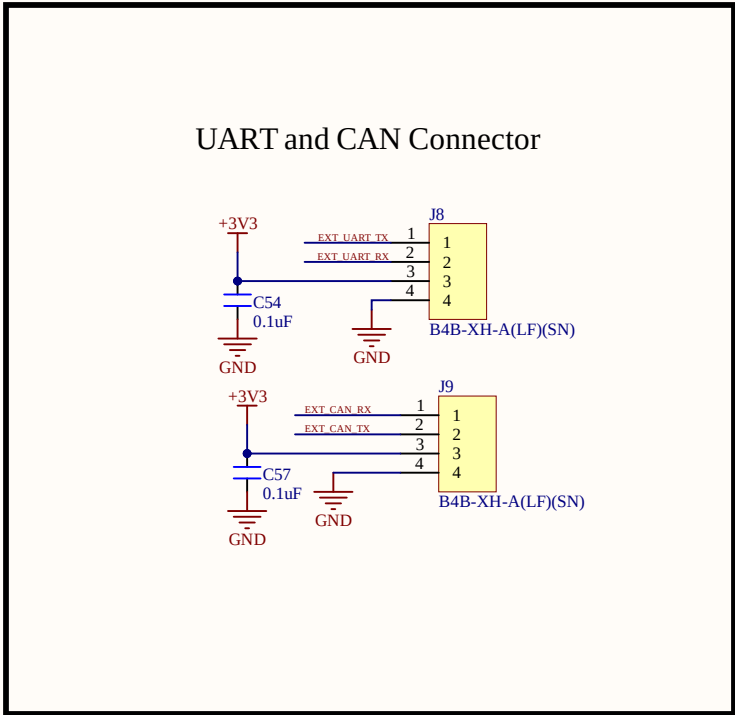
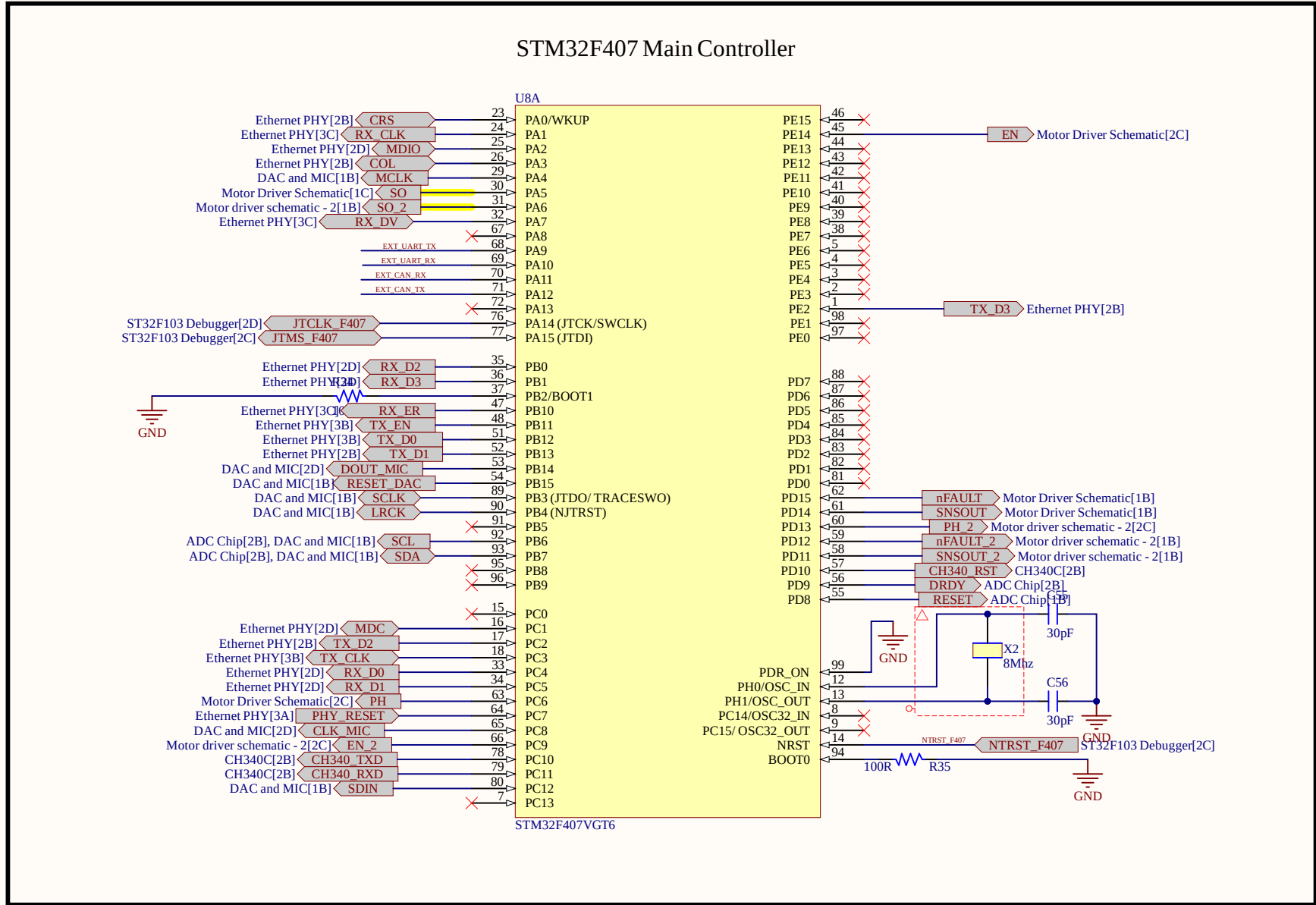
Title **STM32F407 Mixed-Signal Hardware Design**

Size: A4 Number: MSD-001 Sheet 8 of 11 Revision: V1.0

Date: 06-10-2026 Time: 21:29:24 Designer: Maulik Italiya

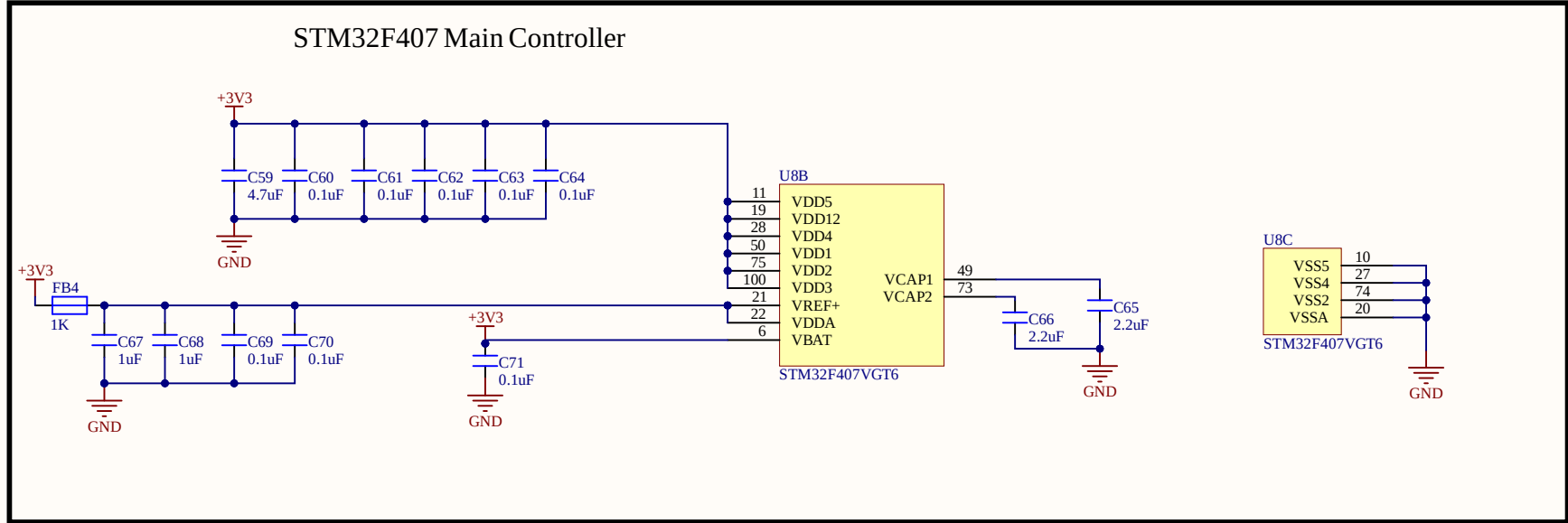
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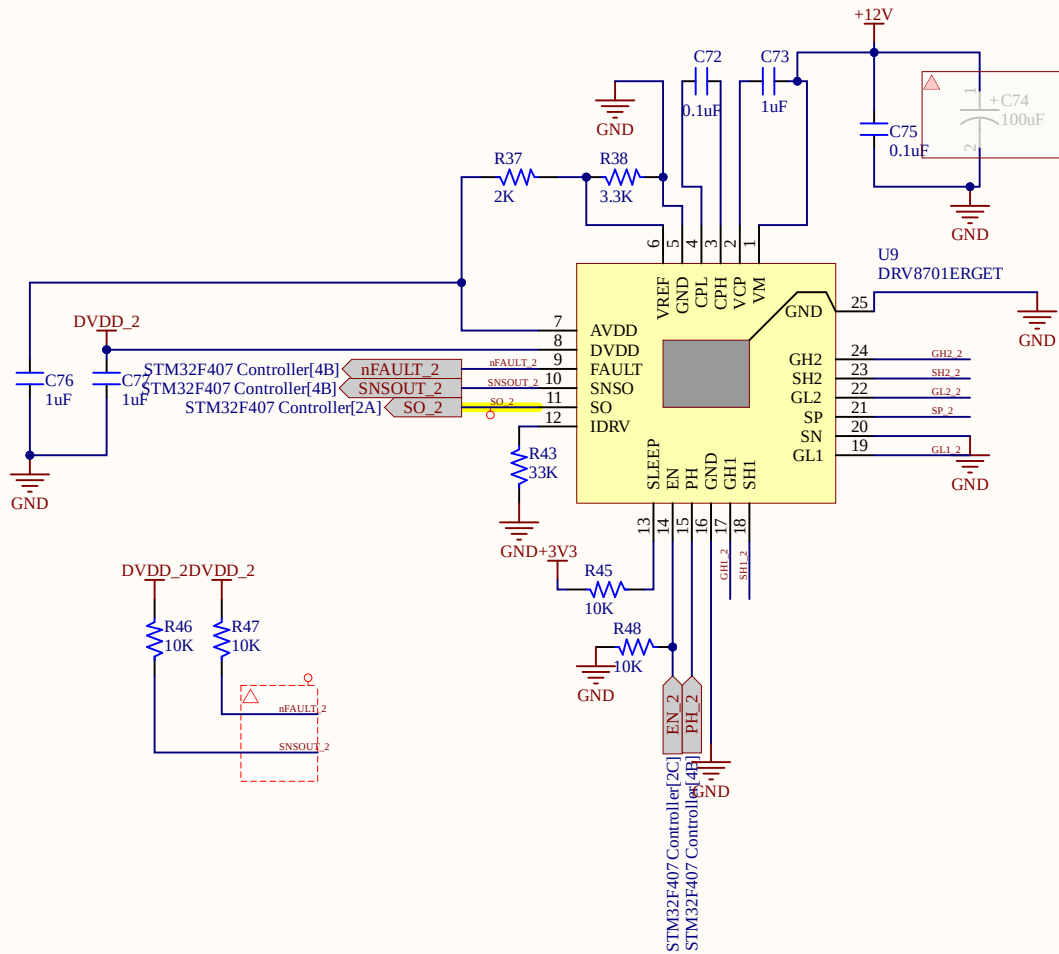
Reset for Main Controller

The diagram illustrates the reset circuit for the main controller. It features a +3V3 supply connected to a 10K resistor (R36). The other end of the resistor is connected to a node labeled NTRST_F40. This node is also connected to a tactile switch (SW3). One terminal of the switch is connected to the NTRST_F40 node, and the other terminal is connected to ground (GND). A 0.1uF capacitor (C58) is connected in parallel with the switch, with one terminal to the NTRST_F40 node and the other to ground.

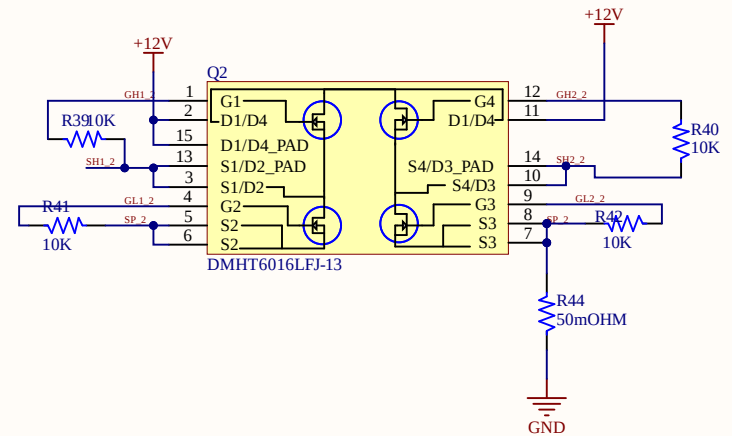


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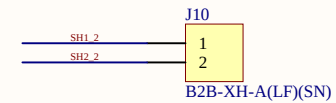
DRV8701 Motor Driver-2



N-MOSFET



Connector for Motor -2



Title STM32F407 Mixed-Signal Hardware Design			
Size: A4	Number: MSD-001	Sheet 10 of 11	Revision: V1.0
Date: 06-10-2026	Time: 21:29:24	Designer Maulik Italiya	
File: D:\Mixed_design_board\My data\Motor driver schematic - 2.SchDoc			

Power Supply Page

